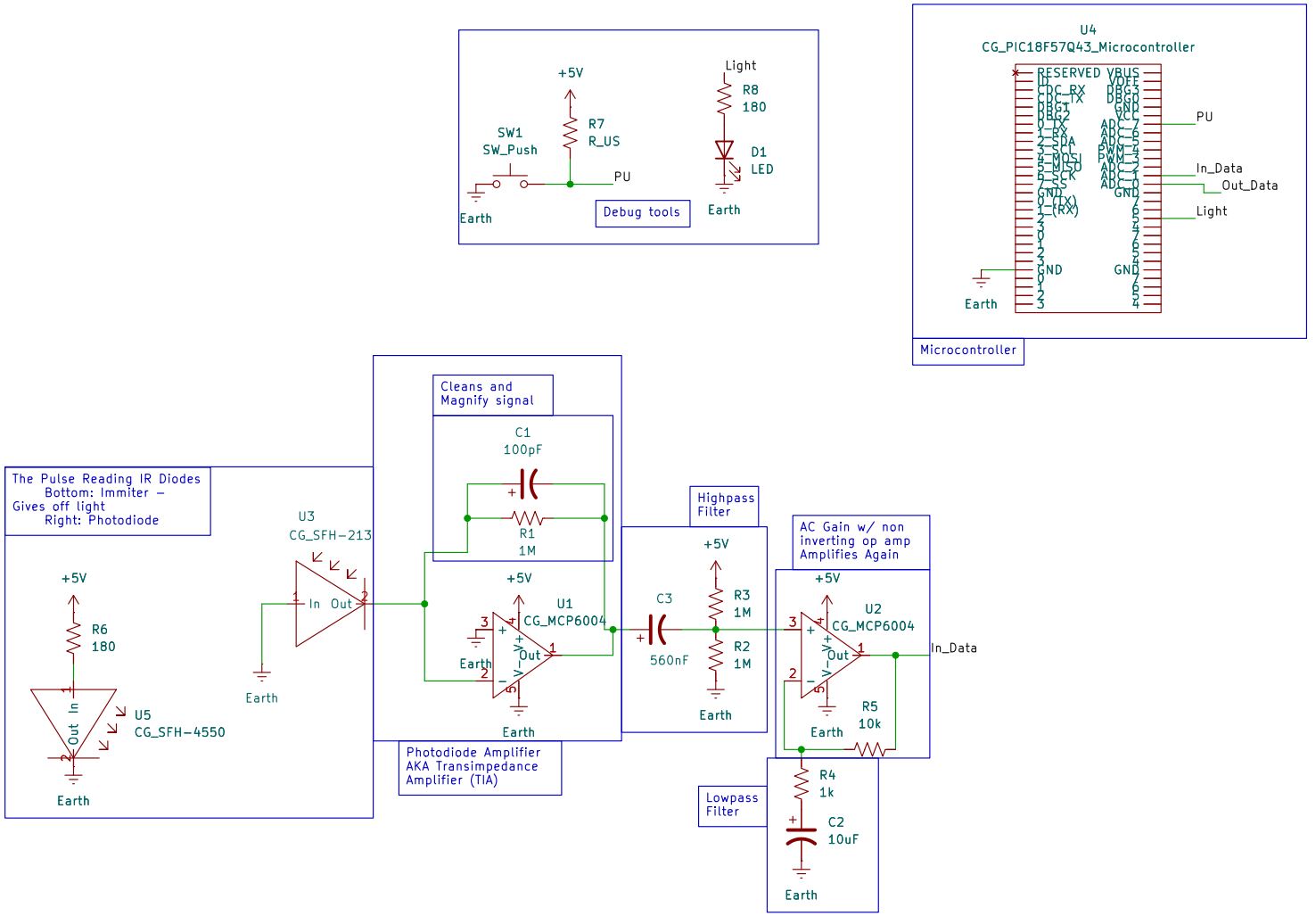

Debug tools
+5V
SW1
SW_Push
R7
R_US
PU
Earth
Light
R8
180
D1
LED
Earth
U4
CG_PIC18F57Q43_Microcontroller
PU
In_Data
Out_Data
Light
Earth
Microcontroller
The Pulse Reading IR Diodes
Bottom: Immiter –
Gives off light
Right: Photodiode
+5V
R6
180
U5
CG_SFH-4550
In
Out
Earth
U3
CG_SFH-213
In
Out
Earth
Cleans and
Magnify signal
C1
100pF
R1
1M
+5V
U1
CG_MCP6004
Out
Earth
Earth
Photodiode Amplifier
AKA Transimpedance
Amplifier (TIA)
Highpass
Filter
C3
560nF
+5V
R3
1M
R2
1M
Earth
AC Gain w/ non
inverting op amp
Amplifies Again
+5V
U2
CG_MCP6004
Out
In_Data
R5
10k
Earth
Lowpass
Filter
R4
1k
C2
10uF
Earth
Sheet: /
File: HRT Rate Monitor Project.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.5
Id: 1/1